AEAC SUAS 2027 Assessment Preview:

Wildlife Monitoring

The Hooved Ecological Resources Division assists with the conservation of ungulate wildlife in the Medicine Hat area and is seeking a UAS solution for wildlife tracking and surveying activities.

The mission proof-of-concept includes multiple challenges:

- Task 1: Herd Survey
- Task 2: Tag, Track, and Sample

2027 Wildlife Monitoring



Task 1: Herd Survey

Coordinate with other search UAVs to detect, count and identify herd animals.



Task 2: Tag, Track, and Sample

Deploy a custom tracker to an animal target and trace its path, while also collecting samples of eggs and/or animal scat.

Task 1: Herd Survey

- Long-distance flight
- Visual survey
- Aircraft avoidance



Task 1: Herd Survey

Teams coordinate with other (simulated) UAVs to survey wildlife in a specified area.

Goals:

- Fly to an animal herd at a remote location.
- Locate, identify, and count animals within a given search boundary.
- Share the airspace with simulated cooperating UAVs – adjust trajectories and communicate with operators.
- Publish and read telemetry data from the competition-provided web server.

Scoring:

- Flight speed to target site
- Correct count and description of wildlife
- Simulated airspace deconfliction
- Safe landing at flightline

Task 2: Tag, Track, and Sample

- Tracker design
- Animal tagging
- Biological sampling



Task 2: Tag, Track, and Sample

Teams track a target animal through the field and collect biological samples from the ground.

Goals:

- Design a custom tracking device (limited mass and size; hook and loop attachment) and tag a target animal with it.
- Collect samples of (simulated!!) animal scat and/or eggs from the ground.
- Using the tracking device, track the target animal as it moves through the flight area and produce a map of its movements.

Scoring:

- Precision of tracker placement on animal target
- Accuracy of animal movement map
- Number and quality/variety of samples collected
- Safe landing at flightline

Telemetry Web Server

This year, a web server is being implemented for telemetry data exchange. Teams will be responsible for ensuring their own internet access at the flight line.

Connection to the server must meet the following requirements:

Transmit: Send telemetry data at 1 Hz and event-based data on demand.

Receive: Process 1 Hz traffic data stream.

Information about the web server and how to connect with it will be provided in the CONOPS in September.

Additional Information

- Only one aircraft is permitted for each task.
- Different aircraft may be used for the two tasks.
- Each UAV must each weigh below 15kg, including payload, at all times during the operation.
- Batteries may be swapped at the flight line during Task 2 but not Task 1.

Important Notes

This Assessment Preview is intended to give preliminary information to help teams budget for equipment this summer.

The full CONOPS will be released in September. Details WILL change!

Please hold your questions on task details until the full CONOPS is released.

If your team has questions about what equipment will be provided, they may be directed to competition@aerialevolution.ca.